

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION
NUMBER

BIOLOGY

Paper 3

9648/03

16 September 2014

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you hand in.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The number of marks is given in brackets [] at the end of each question or part question.

Section A

Answer all questions.

Write your answers in this booklet using dark blue or black pen.

Section B

Write your answers in this booklet using dark blue or black pen.
For Free Response Question, write your answers on the separate Answer Paper using dark blue or black pen.

For Examiner's Use	
1	/ 7
2	/ 8
3	/ 11
4	/ 14
5	/ 12
6	/ 20
TOTAL	/ 72

This document consists of **14** printed pages.

[Turn over

Section A

Answer all questions.

- 1 Restriction fragment length polymorphism, RFLP, refers to differences in fragment length generated by restriction enzymes. These differences are used to detect variations in genomes.

Fig. 1.1 below shows the inheritance of a common human trait which is linked to a restriction fragment length polymorphism (RFLP). Genomic DNA from each family member was assayed for a linked restriction fragment length polymorphism (RFLP). EcoRI sites on Southern blots using probes derived from the gene was shown.

Use the pedigree from the following family along with the Southern blot analyzing inheritance of the RFLP to answer the following questions.

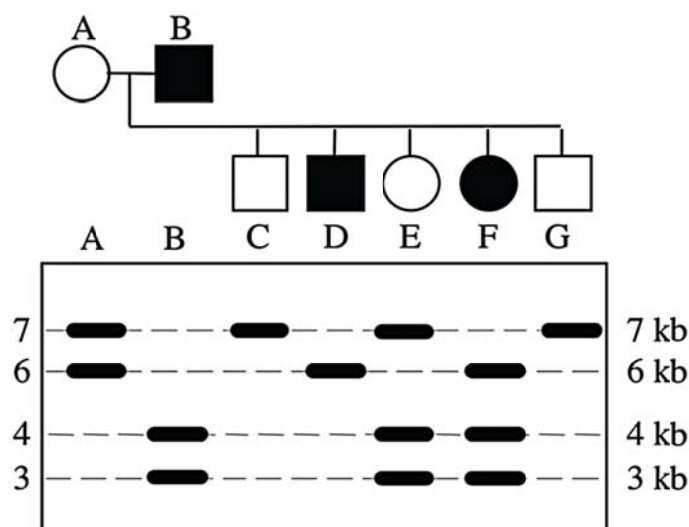


Fig. 1.1

- (a) What is the mode of inheritance for this trait?

[1]

X-linked recessive

- (b) Sketch a suitable diagram to illustrate the region of genomic DNA recognised by the probe in individual A and F. Show the positions of EcoRI sites and the distance between them in kilobases (kb).

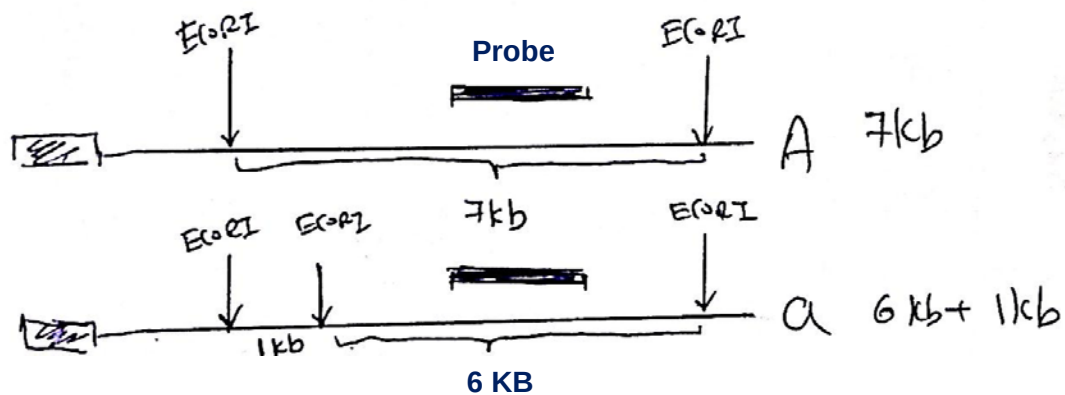
A-Refers to dominant allele (Normal), a- Refers to recessive allele (Disease allele)
Diagram below shows all the 3 different alleles.

Positions of EcoRI sites and where the probes bind to for all the 3 different alleles.

[2]

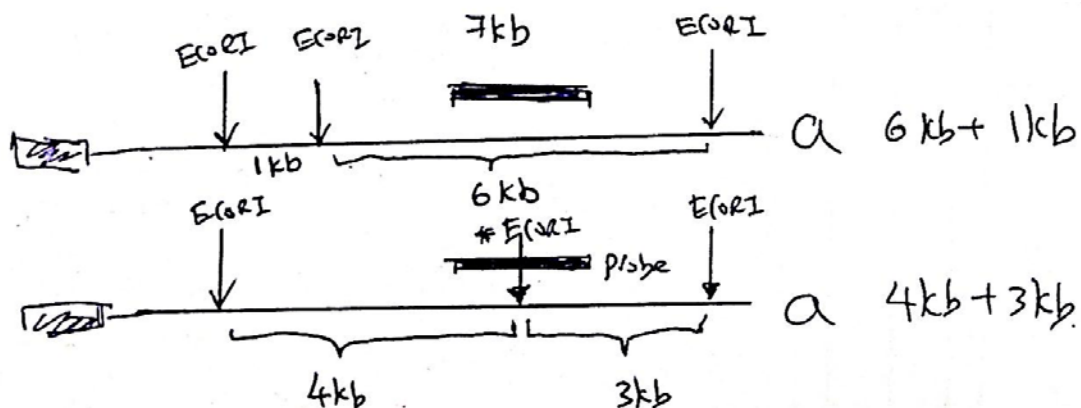
[Note: There should be 2 alleles since each individual is a diploid. Since the probe detects both alleles, diagram should include both allele 1 and 2.]

Individual A: 1 normal allele 1 recessive allele. [7 kb and a 6 kb band on southern blot]



* Note, only 6 KB can be seen from the southern blot, as the probe can bind to 6 kb band, not the 1 kb band.

Individual F: 2 diseased allele [6 kb, 4 kb and 3 kb]



* Note, Both 6 KB, 4 kb + 3 kb can be seen from the southern blot, as the probes can bind to these regions.

- (c) A different study was carried out on another family. Fig. 1.2 below shows a Southern blot with an RFLP detected by using a probe for a sequence located on a sex chromosome. The variant exhibits two forms, one at 7 kb and one at 6 kb. The mother, father and daughter are normal.

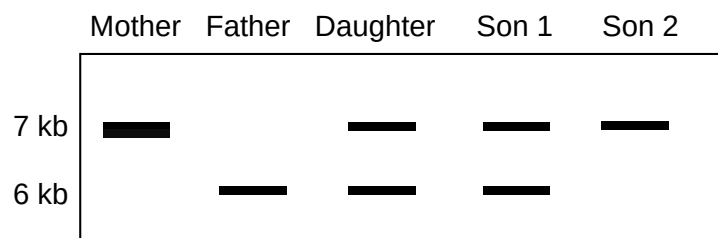


Fig. 1.2

(i) Which son is abnormal?

Son 1.

[1]

(ii) Explain how you arrived at your answer.

- Son 1 is a case of XXY/ Klinefelter's syndrome
- 7 kb band found on an X chromosome, as it is same as mother's 7 kb fragment
- 6 kb band found on X chromosome from father, as it is same as father's 6 kb fragment.

[Y chromosome is not shown because the probe detects only the X chromosome]

[1]

Variable number tandem repeats (VNTRs) are repetitive, non-coding sections of DNA. A particular VNTR is located at the same locus in different individuals, but the number of repeats in that VNTR varies between individuals.

(d) Explain how, in the process of genetic fingerprinting, gel electrophoresis is able to distinguish between the VNTRs that occur at the same loci of different individuals.

1. VNTRs with more repeats are, **longer / greater mass** (or reverse argument)
2. phosphate groups (of DNA) give negative charge ;
3. fragments / DNA, attracted to, anode / positive electrode ;
4. Shorter / lower mass / fewer repeat, pieces move, faster / further in unit time;

[2]

[Total: 7 m]

- 2 The artificial plasmid, pBR322, was constructed to act as a vector. It has often been used to insert human genes, such as the human insulin gene, into the bacterium, *Escherichia coli*.

The plasmid was constructed to include two genes, each giving resistance to a different antibiotic: an ampicillin resistance gene and a tetracycline resistance gene. The plasmid also has a target site for the restriction enzyme, BamHI, in the middle of the tetracycline resistance gene.

A pBR322 plasmid was cut using BamHI and the cDNA gene for human insulin inserted into it. Fig. 2.1 shows pBR322 and the recombinant plasmid.

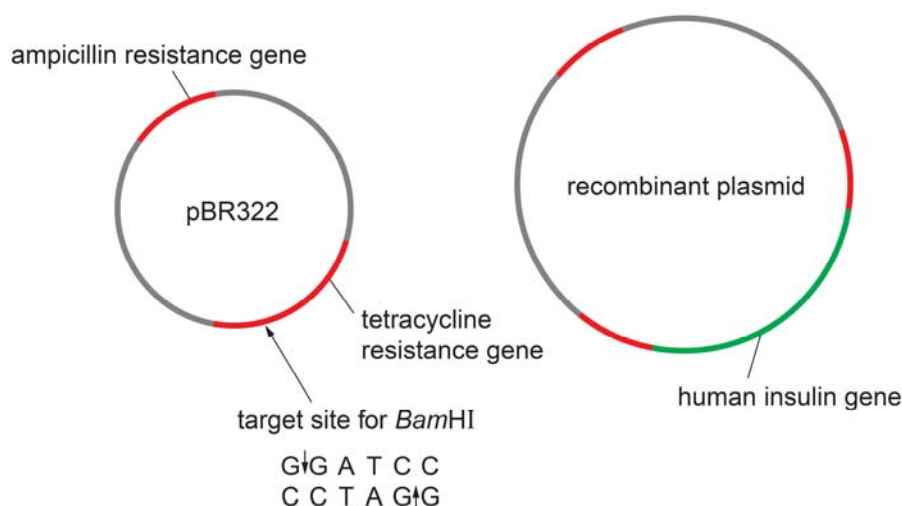


Fig. 2.1

- (a) With reference to Fig. 2.1, describe how a cDNA human insulin gene can be inserted into pBR322 that has been cut by BamHI.

1. **Linker DNA (a short DNA fragment) with Restriction sites (GATC sequence)** are added with **T4 ligase** to both ends of the cDNA of the human insulin gene. – To create complimentary sticky end
2. presence of sticky ends GATC and CTAG on pBR322 ; complementary bases (pairing) ; A to T and C to G ;
3. H-bonds (to sticky ends of plasmid) established with that of cDNA;
4. sugar-phosphate back bone being created; e.g. **formation of phosphodiester bonds** / by **DNA Ligase**

[3]

- (b) Bacteria were then mixed with the recombinant plasmids. Those bacteria which had successfully taken up recombinant plasmids were identified using the following steps:

step 1 – the bacteria were spread onto culture plates containing nutrient agar and ampicillin and incubated to allow colonies to form

step 2 – some bacteria from each of the colonies growing on these plates were transferred to plates containing nutrient agar and tetracycline, as shown in Fig. 2.2.

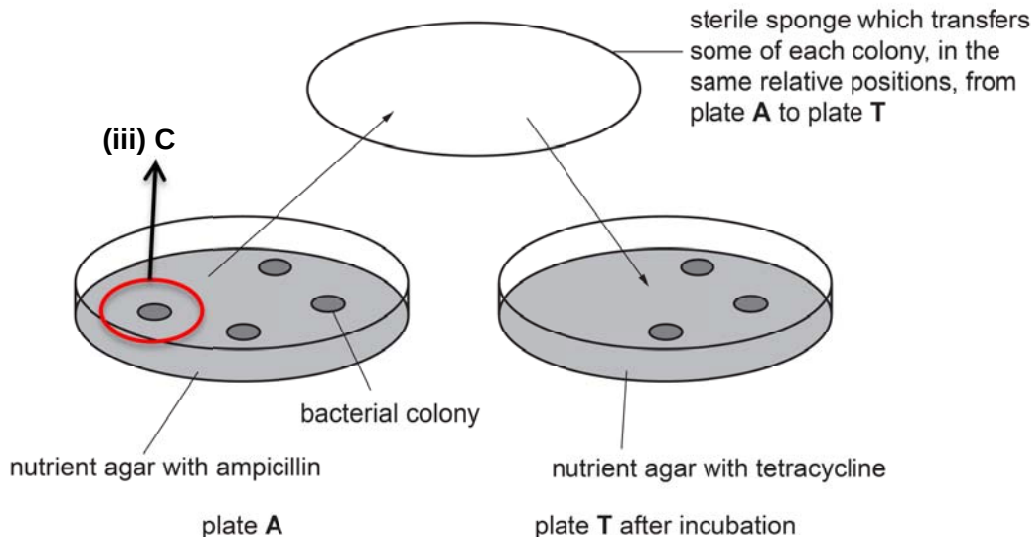


Fig. 2.2

(i) Explain why the bacteria were first spread onto plates containing ampicillin.

1. The idea of **identifying bacteria that, are transformed** / have taken up plasmid / have taken up ampicillin resistance gene ;
2. these bacteria have survived ; these bacteria may contain **pBR322 or recombinant** plasmid / plasmids taken up may not contain human insulin gene ; other bacteria have been killed ;

[1]

(ii) With reference to Fig. 2.1, explain why it is important for *Bam*HI target site to be within tetracycline resistance gene.

1. (BamHI) **disrupts the tetracycline resistance** gene / Insertional inactivation
2. (inserting human insulin gene) makes tetracycline resistance gene inactive ;
3. colonies that are ampicillin-resistant but not tetracycline-resistant have taken up recombinant plasmid / insulin gene ;
4. colonies that survive on, tetracycline / both ampicillin and tetracycline / plate T, have not taken up the recombinant plasmid / insulin gene ; [3 max]

[2]

(iii) Use a labelled line and the letter **C** to identify, on Fig. 2.2, a colony of bacteria that contain the recombinant plasmid.

[1]

Answer on Fig. 2.2 left hand colony on plate A ; [1]

(iv) Plasmid vectors carrying antibiotic resistance genes are now rarely used in gene technology. Explain why antibiotic resistance genes are now rarely used.

1. plasmids (easily) transferred between bacteria of the same species or different species. (Reject Transfer to human cells!!) ;
2. bacteria can acquire antibiotic resistance / renders antibiotic useless
3. Availability of other techniques, which is more accurate and time saving. Unlikely to give rise to false positive results. e.g. Use of radioactively labelled probes etc.

[1]

[Total: 8 m]

- 3 (a) One of the treatments for cystic fibrosis (CF) is to inhale a mist of sterile hypertonic saline (an intensely salty solution). Explain how hypertonic saline can be an effective treatment for CF.

- Hypertonic saline inhalation increases ion concentrations in the airway surface liquid;
- and osmotically draws water from epithelial cells into the airway lumen due to a lower water potential in the airway compared to the cytoplasm of the epithelial cells, thereby diluting mucus;

[2]

- (b) Gene therapy involves transferring normal CFTR alleles into lung epithelial cells with defective genes. Describe how a liposome can be used to insert the normal CFTR allele into a CF patient.

- Liposomes carrying normal CFTR allele is sprayed into the nose and mouth of CF patients in the form of an aerosol spray
- Liposomes fuse with the cell membrane of epithelial cells, releasing the normal CF allele into the cytoplasm of cells

[2]

- (c) “Gene therapy for CF does more than treating the symptoms of the genetic disease; it targets the cause of the disease.”

Explain why this statement is true.

- Gene therapy involves delivery of normal/functional CFTR allele into the human body (by the means of a vector), hence targeting the cause CF as it enable the formation of functional CFTR

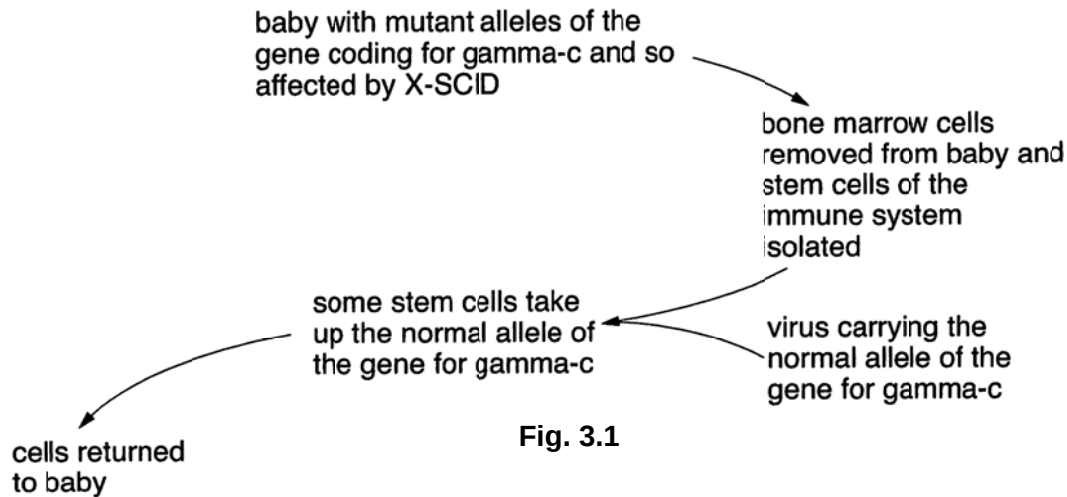
[1]

- (d) Explain why it is easier to perform gene therapy on recessively inherited diseases.

1. recessively inherited disease is caused by the presence of two recessive alleles
2. only 1 normal allele needs to be inserted since it is dominant and not masked
3. no need to remove/inactivate the recessive/mutant allele or For the case of dominant mutation, it is harder to inactivate.

[3]

Gene therapy for several types of SCIDs depends on the isolation of stem cells of the immune system. X chromosome-linked SCID (X-SCID) is caused by an inability to produce a protein, gamma-c. The procedure for the gene therapy is shown in Fig. 3.1.



- (e) With reference to the procedure shown in Fig. 3.1,
- (i) suggest the type and potency of stem cells used in this treatment.

Hematopoietic / blood / adult stem cells & multipotent (To get full credit)

[1]

- (ii) describe the properties of the stem cells that are used.

- capable of differentiating into specialized cells / cells of related functions (e.g. lymphocytes, red blood cells, platelets)
- no tissue-specific structures that allow specialized functions, unspecialized in structure and function
- capable of long-term self-renewal / continually divide for long period of time (*Reject: capable of mitosis to form more cells*)

[2]

[Total Marks: 11]

Name: _____ Class: 2bi2___ / 2IPbi2___

26

Section B

Answer **all** the questions in this section.

- 4 Transgenic plants over-expressing the vacuolar H^+ -pyrophosphatase are much more resistant to high concentrations of NaCl and to water deprivation than the wild-type strains. These transgenic plants accumulate more Na^+ and K^+ in their leaf tissue than the wild type. Vacuolar H^+ -pyrophosphatase of Arabidopsis is encoded by a single gene, *AVP1*. In an experiment, agrobacterium-mediated transformation was performed on Arabidopsis thaliana to produce eight transgenic lines, with enhanced salt tolerance. The procedure of culturing cells already transfected with *AVP1* gene can be divided into four stages.

- (a) (i) Briefly describe the main idea of Stage 2 and 3.

Stage 1: Establishment of aseptic culture.

Explant is disinfected and grown in suitable medium containing nutrient and agar.

Stage 2: Multiplication and axillary shoot enhancement

Formation of callus by adding equal ratio of auxin and cytokinin to the nutrient medium.

Stimulate shoot formation by placing call in growth medium with auxin < cytokinin.

[1]

Stage 3: Rooting

Shoots are then recultured into a growth medium with auxin > cytokinin to induce root formation.

[1]

Stage 4: Acclimatisation

Plantlets transferred to soil in sheltered environment. Young plants are eventually replanted in an open uncontrolled environment.

- (ii) Identify one issue that can occur in stage 4.

- infectious disease in soil
- desiccation as plants are now growing in an open environment

[1]

- (b) Two of the cell lines (AVP1-1 and AVP1-2) were chosen for further analysis, along with the wild type.

The three cell lines were placed under a water-deficit condition for eight days. And results of their relative water content of wild type and AVP1 transgenic plants were recorded in Fig. 4.1

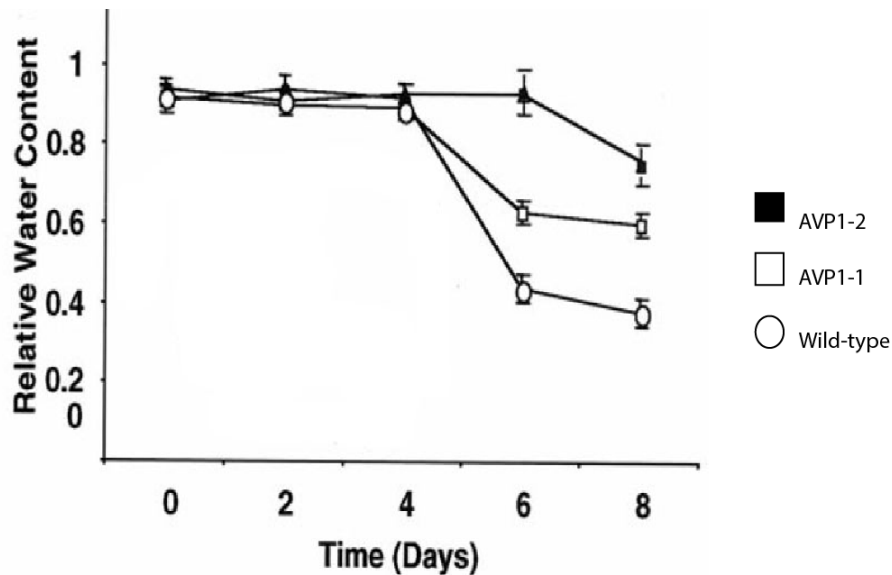


Fig. 4.1

- (i) With reference to Fig. 4.1, explain the effect of the transgene on the plantlets and drought-resistance.

- Relative water content for all three cell lines stayed the same from day 0 to 4, but decreases for all three cell lines from day 4 to day 8 / Ref: to general trend

Any 2 points below 2 marks

- Wild-type showed the sharpest decrease from 0.9 to 0.4 rwc
- AVP1-1 to 0.6 rwc
- AVP1-2 to 0.8 rwc

AND

- AVP1-1 and AVP1-2 are able to retain water content better than wild-type, thus more resistant to drought

[4]

- (ii) Although drought-resistant plants can provide benefits to farmers in arid regions, they may also have detrimental effects to the environment. Suggest one such effect.

- Transfer of the AVP1 gene to weeds
- Reduce genetic diversity as plants carrying AVP1 will be at a selective advantage c.f. to plants without

[1]

- (c) Further research into *AVP1-1* and *AVP1-2* cell line showed that there is a distinct genetic difference between them, and both are tagged with a different genetic marker; *AVP1-1* with genetic marker 1 and *AVP1-2* with genetic marker 2.

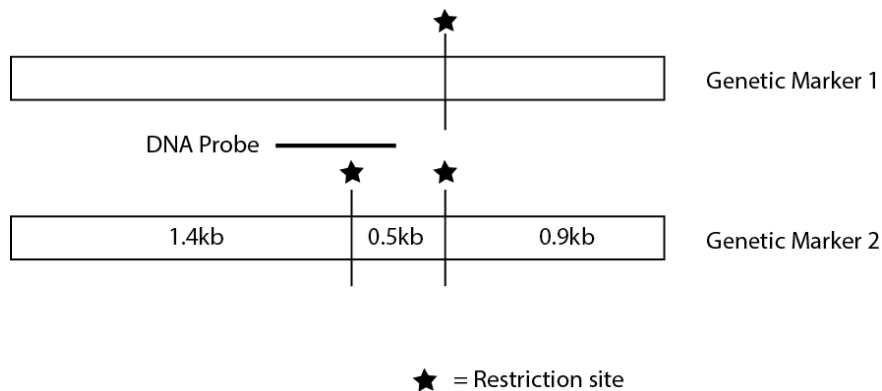


Fig. 3.2

(i) Explain how restriction fragment length polymorphism comes about.

- Difference in nucleotide sequences between chromosomes
- causes restriction sites to be created / disappear at a loci

[2]

(ii) With reference to Fig. 3.2, describe how RFLP can be used to identify cell line carrying *AVP1-2* gene.

Max. 1 mark

- Difference in DNA sequence of between the two cell lines result in differences in restriction enzyme recognition site and result in different DNA fragments

[4]

Max. 2 marks

- DNA restriction fragments are separated according to size / molecular weight by (agarose) gel electrophoresis
- Smaller fragments migrate faster
- Agarose gel placed in NaOH to separate DNA into single-stranded DNA, transferred onto nitrocellulose blot by capillary action
- Blot is incubated with single-stranded radioactively labeled nucleic acid probe which will hybridise to specific target nucleotide sequences by complementary base pairing
- Autoradiography carried out and X-ray film is exposed to X-ray. Target DNA bands bound to radioactively labeled probes will show up as a black band on the X-ray film

Max. 1 mark

- By comparing the DNA bands in the DNA/genetic fingerprint, one can detect and observe the difference between *AVP1-1* (1.9kb) and *AVP1-2* (1.4 kb and 0.5kb) allele.

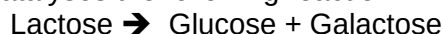
[Total: 14 m]

5 Planning Question

Write your answers to this question on the separate answer paper provided.

Inhibitors are substances that reduce the activity of enzymes. When the inhibitor binds reversibly to the active site of the enzyme, it is known as a competitive inhibitor. Often a competitive inhibitor is a similar shape to the substrate. When a non-competitive inhibitor combines with an enzyme, the shape of the active site is altered sufficiently to prevent the substrate from combining with it.

The enzyme β -galactosidase catalyses the following reaction:



The chemical ONPG (o-nitrophenyl β -D-galactopyranoside) is also degraded by the enzyme, β -galactosidase:



The ONP produced is yellow, allowing the rate of this reaction to be followed colorimetrically. The concentration of ONP is proportionate to its absorbance. Using this information and your own knowledge, design an experiment to determine whether iodine solution, an inhibitor of this reaction, is a competitive or non-competitive inhibitor.

You must use:

- 6 cm³ ONPG stock solution
- β -galactosidase stock solution
- 40 cm³ buffer pH 8
- 5 cm³ I₂/KI solution
- Colorimeter (440 nm filter)

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- cuvettes
- beaker of crushed ice
- droppers
- eye protection
- gloves

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results are accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- Include reference to safety measures to minimize any risks associated with the proposed experiment.

Mark Scheme:**Scientific reasoning / Explanation of theory [3 marks]**

- Ref to basic function of enzyme inhibitors, e.g. enzyme inhibitors affect enzyme activity resulting in a decrease in the rate of enzyme catalyzed reactions
- Ref to competitive inhibition:
 - Competitive inhibitor is structurally similar to the substrate
 - Binds reversibly to the active site of the enzyme and competes for the substrate for active site
 - Effect of inhibition overcome by increasing substrate concentration
- Ref to non-competitive inhibition:
 - does not bind to active site of the enzyme
 - Binds to inhibitor site, e.g. allosteric site
 - Inhibition cannot be overcome by increasing substrate concentration
- Ref to ONPG color / absorbance value

Variables [2 marks]

(1 mark)

- Controlled variables: Temperature, pH, Volume of inhibitor used, with explanation

(Both below present and correct for 1 mark)

- Dependent: absorbance value / color intensity / product formed (ONP)
- Independent: ONPG volume / concentration used

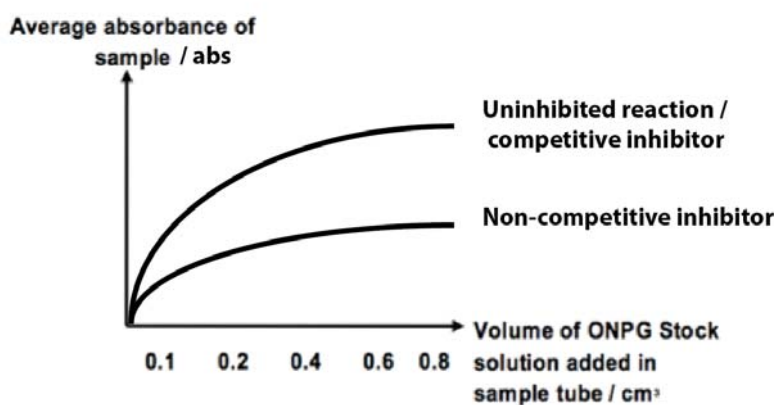
Procedure [4 marks, 1 mark each]

- Using serial dilution / simple dilution to obtain at least 5 known concentration of ONPG, diluting with either buffer solution or distilled water
- Stating that the tube be kept in ice to reduce enzyme activity / maintaining constant temperature during enzymatic reaction
- Stating equal volume of I_2/KI solution used
- Specific time set for reaction (1 – 2 minutes after addition of enzyme)
- Means of measuring 2 – 3 absorbance value for each tube using colorimeter / spectrophotometer, with absorbance value set between 420 – 440 nm
- Stating replicates and repeats to be performed

Data recording [1/2 mark each, max. 2 marks]

- Tabulation of data with headings and units;
- Including average value for absorbance value
- Plot graph of absorbance value (abs) against ONPG concentration / volume
- Correct labeling of x and y – axis

Set without inhibitor					
Cuvette no.	Vol. of ONPG stock solution / cm^3	Absorbance value 1 / abs	Absorbance value 2 / abs	Absorbance value 3 / abs	Average Absorbance / abs
1					
2					
3					
4					
5					
6					
Set with inhibitor					
Cuvette no.	Vol. of ONPG stock solution / cm^3	Absorbance value 1 / abs	Absorbance value 2 / abs	Absorbance value 3 / abs	Average Absorbance / abs
1					
2					
3					
4					
5					
6					



Safety Precautions / Risk Hazards [1 mark]

- take care when handling glassware such as beakers and test tubes
- do not allow reagent such as ONPG to have contact with skin or eyes, wash when in contact
- wear goggles throughout the experiment
- AVP;

Free Response Question

Write your answers to this question on the separate answer paper provided.

Begin each section of the question on a new piece of answer paper.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections (a), (b), etc., as indicated in the question.

- 6 (a) Explain how insulin genes are cloned using *E. coli* cells to produce eukaryotic proteins. [6]
- (b) Compare and contrast between treatment of cystic fibrosis and SCID. [6]
- (c) Explain the factors that prevent gene therapy from becoming an effective treatment for genetic diseases. [8]

--- End of Paper ---

- 6 (a) Explain how insulin genes are cloned using *E. coli* cells to produce eukaryotic proteins. [6]

Isolating mRNA for Insulin: [max 1]

1. Isolate **mRNA for insulin** (from islets of Langerhans of human pancreatic cells as template) and synthesise **cDNA using reverse transcription** / reverse transcriptase to form cDNA.
2. Convert the **single stranded cDNA** to **double stranded DNA** by first removing the mRNA using ribonuclease and then synthesising the complementary strand using DNA polymerase.

Or Isolate Insulin Gene from cDNA library using radioactively labelled probe. Perform PCR using Primers which flank the insulin to amplify the insulin Gene.

A

Formation of recombinant DNA molecules with cDNA and Plasmid: [max 2]

3. **DNA linkers** (containing the restriction site) added to each of the cDNA using (DNA ligase)

or 3.1: Add additional guanine nucleotides to create artificial sticky ends for the insulin DNA using terminal transferase, Cut plasmid vector with restriction enzyme to create blunt ends and add additional cytosine nucleotides to create artificial sticky ends for the plasmid. Mix the cut plasmid and human cDNA to allow hydrogen bonds between complementary bases of the artificial sticky ends.

or 3.2 **digest plasmids** (containing lacZ and amp^R or any antibiotic marker genes) and the **cDNA with the same RE to produce sticky ends**. (reference to plasmids cut within lacZ by RE)

4. mix the cDNA and the plasmids together and let the sticky ends anneal by Hydrogen bonds by complementary base pairing
5. Add **DNA ligase** to covalently join the fragments with phosphodiester bonds (covalent bond to form recombinant plasmids)

B1

B2

Transformation of recombinant DNA into bacteria [max 1]

7. Reference to use of calcium chloride and heat shock method to increase uptake of recombinant plasmid by competent **E.coli** cells. Plasmids containing A and B chain were transformed separately into E.coli;

C

Selection of transformed E.coli and mass production of insulin[max 1]

8. Select for **transformed E.coli colonies that acquire the recombinant** plasmid using selective markers present on the plasmid/ replica plating or blue-white screening on appropriate selection medium (i.e. antibiotic agar plates or agar plate containing X-gal and antibiotic respectively).

D

9. select colonies which are white in colour as these are the ones with recombinant plasmid (Reject: bacteria appears white) due to inactivation of Lac z gene

Mass production and Purification [Max 1]

10. selected bacteria grown (in large quantities) in a suitable medium and bacteria secrete insulin which is extracted, purified and packaged for use

E

11. Cyanogen bromide is used to cleave the methionine residue; Purified A and B chains are attached to each other via disulphide bonds to form functional insulin;

6 (b) Compare and contrast between treatment of cystic fibrosis and SCID.

Similarities (max. 3 marks):

- Both treatments involve the replacement of a defective gene with a normal functional gene, so that the body can make the correct enzyme/protein and consequently eliminate the cause of the disease.
- Both treatments may cause a mild immune response.
- For both treatments, the viral gene delivery system can be used

Differences (max. 3 marks):

	SCID gene therapy	Cystic Fibrosis gene therapy
Target cells	Blood stem cells from bone marrow	Epithelial cells of respiratory tract
Vector used for gene transfer	Treatment involves the use of a retrovirus as a vector	Plasmids (inside cationic liposome) are used to transfer the DNA of the normal functional cystic fibrosis gene into the target cell. OR adenovirus used as a vector
Treatment	Ex vivo – where target cells are removed from patient, retrovirus carrying normal functional ADA gene is introduced, then cells are transferred back into patient.	In vivo – where liposomes with plasmid carrying normal functional CFTR gene is introduced into the patient directly via aerosol spray
Treatment procedure – Entry into target cell	Retrovirus enters the blood stem cell via attachment / adsorption onto target cell membrane, followed by fusion of viral envelope with cell membrane.	Lipid molecules on the liposome bind to cell surface receptors on target cell, and the liposome is engulfed by the membrane. Cationic liposome carrying plasmid with normal function gene enters the target cell via endocytosis.
Treatment procedure – use of RT for expression of normal functional gene	The retrovirus vector particles upon entering the target cells, synthesizes the ADA gene using reverse transcriptase.	The recombinant plasmid with normal gene enters the nucleus, and is directly transcribed and translated to give normal CFTR proteins.
Treatment procedure – long-term expression of normal functional gene	Integration of normal functional gene into blood stem cell DNA enables long-term expression of the normal functional gene.	With no integration of normal functional gene into target cell DNA, there is no long-term expression as normal functional gene may be degraded by nucleases.

- 6 (c) Explain the factors that prevent gene therapy from becoming an effective treatment for genetic diseases. [8]

Difficult to get DNA to integrate into target cell genome [1]

Difficult to get DNA to integrate into target cell genome / low efficiency of gene transfer, hence introduced gene may not segregate equally to daughter cells if the cells are actively dividing

Inserted gene may not expressed [1]

Inserted genes may not be expressed because they may be integrated into a highly condensed heterochromatic region

Insertional mutagenesis [1]

Insertional mutagenesis caused by e.g. retroviruses / most transgenes insert ectopically, hence displacing other genes throughout the genome / may cause disruption of other useful genes / Eg. insert into tumour suppressor genes / proto-oncogenes, leading potentially to cancer

Difficult to control activity of cells [1]

Difficulties in controlling the activity of the transferred gene / cells must make appropriate amounts of the gene product at the right time and in the right place / diseases such as thalassemia require a precise level of expression of the normal gene, which gene therapy cannot achieve

Short-lived nature of gene therapy [1]

- Results can be short-lived / transient expression of the transgene if not integrated into host genome / typically short-lived because of difficulty in integrating normal, functional allele into host cell genome
- Due to rapidly dividing nature of many cells
- Natural death of treated cells means that new cells need to be constantly transformed
- Patients will have to undergo multiple rounds of gene therapy;

Immune response [any 1, 1m]

Introduced vectors may stimulate the immune system and render viral vectors useless / gene therapy using common viruses will be difficult since patients would have acquired antibodies during the previously infection / if virus vector triggers the immune response the first time, the immune system will mount a stronger and faster response the second time rendering the therapy ineffective subsequently

Most viruses modified for use as gene delivery vectors are naturally pathogenic and disease-causing / fear that the viral gene delivery vector, once introduced inside the patient, may recover its ability to cause disease / viral vector may regain virulence, and these may still cause inflammatory responses in the recipients, in some cases

The introduction of a foreign DNA into the human tissues risks eliciting a powerful immune response by both the T-cells and B-cells, resulting in the production of antibodies that would destroy the foreign DNA, if it is introduced or administered again;

Multigene disorders [any 1, 1m]

Gene therapy is useful for treating conditions or disorders that arise from mutations in a single gene / difficulty in treating multigene disorders, because difficult/impossible to introduce many genes at the same time

Many common diseases, (name one eg. heart disease, high blood pressure, Alzheimer's disease, arthritis, and diabetes) are caused by the combined effects of variations in many genes

Presence of dominant alleles [1m]

Difficulty in correcting a disease which is due to the presence of a dominant allele

Problems with vectors [any 1, 1m]

Difficulty in finding suitable vectors for large genes / viral vectors can only accept genetic material of a certain size / cannot find a suitable viral vector due to viruses' host specificity

Not applicable to non-dividing cells [1m]

Not applicable to non-dividing cells / if the modified target cell (with the normal functional allele) does not divide and multiply, the introduced gene is lost when the cell dies.